Drepto MenstroHerb Sheet Summary

INTRODUCTION:

Drepto MenstroHerb Sheet is a herbal transdermal patch indicated for the management of menstrual pain. It is a product developed by Drepto Biodevices Private Limited.

DESCRIPTION:

Menstrual pain is a common gynaecological condition characterized by painful menstruation, typically presenting as cramping pain in the lower abdomen that may radiate to the lower back, thighs and may be associated with nausea, vomiting, headache, fatigue, or diarrhoea. The pain occurs in the absence of any pelvic pathology and is mainly attributed to increased prostaglandin F2α production, which stimulates uterine contractions and vasoconstriction, resulting in reduced uterine blood flow, hypoxia, and pain. Menstrual pain is highly prevalent, affecting approximately 50–90% of menstruating women worldwide, with moderate to severe symptoms reported in about 10–25% of cases. The condition is a leading cause of school and work absenteeism among adolescent girls and women of reproductive age, significantly impacting quality of life, daily functioning, and productivity.

In view of this, Drepto Biodevices Private Limited has come up with herbal transdermal patch under the brand name “Drepto MenstroHerb Sheet” in the management of painful cramps associated with menstruation.

Drepto MenstroHerb Sheet is a herbal transdermal patch with a composition comprising extracts of *Curcuma longa* (*Haridra satva*), *Mentha piperita* (*Peppermint satva*), *Boswellia serrata* (*Shalaki Guggul*), and *Eucalyptus globulus* (*Taila Parni*). In the sheet, each herbal ingredient contributes through a complementary mechanism to relieve menstrual pain and associated inflammation. *Curcuma longa* (*Haridra satva*) exerts potent anti-inflammatory and analgesic effects primarily by inhibiting cyclooxygenase (COX) and lipoxygenase pathways, thereby reducing prostaglandin synthesis responsible for uterine cramps. *Mentha piperita* (*Peppermint satva*) provides antispasmodic and cooling effects due to menthol, which relaxes uterine smooth muscle by modulating calcium channels and offers rapid pain relief and soothing action at the site of application. *Boswellia serrata* (*Shalaki Guggul*) acts as a strong anti-inflammatory agent by inhibiting 5-lipoxygenase and leukotriene synthesis, thereby reducing pelvic inflammation and chronic pain associated with menstrual disorders. *Eucalyptus globulus* (*Taila Parni*) contributes analgesic, anti-inflammatory, and mild counter-irritant effects, improving local blood circulation and enhancing transdermal penetration of actives. It also contains cinnamon oil which has cinnamaldehyde, thereby providing antispasmodic, anti-inflammatory effects by relaxing uterine smooth muscles, reducing prostaglandins, and suppressing cytokines to alleviate menstrual pain. Together, these ingredients, delivered via a transdermal patch, provide sustained, localized relief from menstrual pain while minimizing systemic side effects.

Patent filing Details: Drepto Biodevices Private Limited has filed an Indian patent application **(IN202521129798)** dated Dec 22, 2025 for their said product invention entitled “Transdermal Herbal Patch Composition for Management of Menstrual Pain”. This enables Drepto Biodevices Private Limited to have uniqueness and exclusivity of its product in the market.

Trademark Filing Details:

Drepto Biodevices Private Limited has also filed a **trademark application (7344054)** for the brand name “MENSTROHERB SHEET” dated Nov 15, 2025. This enables Drepto Biodevices private limited to have brand exclusivity in the market.

Conclusion:

Therefore, the product MenstroHerb Sheet, developed by Drepto Biodevices Private Limited, provides safe, effective, and sustained relief from menstrual pain by targeting inflammation, uterine muscle spasm, and local discomfort through a synergistic herbal transdermal delivery system, while minimizing systemic side effects and improving patient comfort and compliance.



FINAL STUDY REPORT

STUDY TITLE

SKIN IRRITATION TEST OF TEST ITEM "DREPTO MENSTROHERB SHEET" WITH NEW ZEALAND WHITE RABBITS

TEST METHOD: ISO 10993-23:2021 / Amd 1:2025

STUDY No.: NLS-TX-270/26

STUDY CODE: SIT-RB

STUDY COMPLETION DATE: 02/09/2026

STUDY DIRECTOR AND AUTHOR

Rohith Kumar H.N.

SPONSOR

DREPTO BIODEVICES PVT. LTD
OFF. SINE IIT, BOMBAY, 1001-T1, RUSTOMJEE OZONE
CO-OP HSG, BEH.INORBIT, GOREGOAN (MUMBAI)
MUMBAI, GOREGOAN WEST, MAHARASHTRA
INDIA, 400104

TEST FACILITY

NINKILIM LIFE SCIENCES PRIVATE LIMITED
PLOT No. 553P-14, VASANTHANARSAPURA
2nd PHASE, KIADB INDUSTRIAL AREA
TUMAKURU-572128
KARNATAKA
INDIA

STATEMENT OF CONFIDENTIALITY

This report contains confidential and proprietary information of "Drepto Biodevices Pvt. Ltd, off. Sine IIT, Bombay, 1001-T1, Rustomjee Ozone. Co-op HSG, Beh.Inorbit, Goregoan (Mumbai) Mumbai, Goregoan West, Maharashtra, India, 400104" which will not be disclosed to anyone except the employees of this company or to persons authorized by law or judicial judgment without the expressed or written approval from Drepto Biodevices Pvt. Ltd.

COMPLIANCE STATEMENT

This Study No.NLS-TX-270/26 was performed in compliance with the following:

- ISO/IEC 17025:2017, General Requirements for the Competence of Testing and Calibration Laboratories.
- OECD Principles of GLP for testing of chemicals as specified by international [(C (97)186/Final] legislation.
- The Study Plan was mutually agreed by Sponsor on 11/08/2026 and approved by the Study Director on 12/08/2026.
- This study was conducted in accordance with the Standard Operating Procedures (SOP's) of Test Facility.

DECLARATION

This is to declare that the work was performed under the supervision of the Study Director and in accordance with the described procedures. It is assured that the reported results faithfully represent the raw data obtained during the experimental work. No circumstances have been left unreported which may have affected the quality or integrity of the data or which might have a potential bearing on the validity and reproducibility of this study.

The Study Director accepts overall responsibility for the technical conduct of the study as well as the interpretation, analysis, documentation and reporting of the results.

02/09/2026

Rohith Kumar H.N.
Study Director
Sign. and Date

STATEMENT OF QUALITY ASSURANCE UNIT

The Study No. NLS-TX-270/26 entitled "Skin Irritation Test of Test Item "Drepto Menstroherb Sheet" with New Zealand White Rabbits" has been inspected in compliance with the ISO/IEC 17025:2017 and OECD Principles of GLP.

This study was inspected and the findings were reported to the Test Facility Management (TFM) and Study Director (SD) on the dates shown below:

Inspection Date	Inspection Phase	Reporting Date
Initiation Phase		
07/08/2026	Study Plan -Draft Review	07/08/2026
12/08/2026	Study Plan - Final Review	12/08/2026
In-Life Phase		
18/08/2026	Body Weight & Test Item Application	18/08/2026
19/08/2026	Skin Reactions Scoring	19/08/2026
Reporting Phase		
24/08/2026	Study Report -Draft Review	24/08/2026
02/09/2026	Study Report - Final Review	02/09/2026

Inspections were performed according to the Standard Operating Procedures (SOP's) of the Test Facility's Quality Assurance Unit (QAU). This report was inspected against the approved Study Plan and pertinent raw data and accurately reflects the raw data.

Keerthi Kumar RH
02/09/2026

Keerthi Kumar R.H.
Quality Assurance Unit
Sign. and Date

STATEMENT OF TEST FACILITY MANAGEMENT

Study Number : NLS-TX-270/26

Study Title : Skin Irritation Test of Test Item "Drepto Menstroherb Sheet" with New Zealand White Rabbits.

This is to affirm that for the above-mentioned study, Test Facility Management (TFM) has made available all the resources to the Study Director necessary for conduct of the study in compliance with ISO/IEC 17025:2017, OECD principles of GLP and mutually agreed Study Plan.

02/09/2026

Jyothi G.
Test Facility Management
Sign. and Date

ABBREVIATIONS AND SYMBOLS

Amd	:	Amendment
CCSEA	:	Committee for Control and Supervision of Experiments on Animals
Cm	:	Centimetre
Cm^2	:	Square centimeter
dB	:	Decibel
GLP	:	Good Laboratory Practice
i.e.	:	that is
IAEC	:	Institutional Animal Ethics Committee
IEC	:	International Electrotechnical Commission
ISO	:	International Organization for Standardization
Kg	:	Kilogram
NLS	:	Ninkilim Life Sciences Private Limited
Min	:	Minute
mg	:	Milligram
mL	:	Milliliter
mm	:	millimeter
No.	:	Number
OECD	:	Organisation for Economic Co-operation and Development
RO	:	Reverse Osmosis
Sign.	:	Signature
SIT-RB	:	Skin Irritation Test in New Zealand White Rabbits
TIDS	:	Test Item Information Data Sheet
TX	:	Toxicology
ULR	:	Unique Lab Report
%	:	Percentage
&	:	And
°C	:	Degree Celsius
<	:	Less than

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1. STUDY SUMMARY

The objective of conducting this *In-Vivo* toxicity study / test was to evaluate the irritancy potential of the Test Item "Drepto Menstroherb Sheet" following single exposure topical application onto the intact skin of the Rabbit to evaluate the possibility of skin reaction (Erythema and Oedema). This test also provides information on health hazards that were likely to arise from acute exposure by the intended clinical route in humans. This test was conducted in accordance with Test Method ISO 10993-23:2021-Biological evaluation of medical devices - Part 23: Tests for Irritation / Amd 1:2025.

Four Female Rabbits were acclimatized for 5 days and three Rabbits were used for the application. Prior to the test item application (i.e., 18 hours and 11 minutes), the fur on the dorsal region of the animals were closely clipped on both side (right and left side) of the spinal column (10 cm x 15 cm=150 cm^2) using electric hair clipper. Care was taken to avoid abrasion of the skin and there was no skin abrasion found on any of the animals. Animals with healthy intact skin were used for the test. The unused one rabbit was euthanized by sodium thiopental injection on Day 1 post-application.

For control application sites, the filter paper was cut into size of 6.25 cm^2 (2.5 cm x 2.5 cm) in a Laminar Air Flow Cabinet prior to application and was impregnated with RO water and applied onto the clipped surface of the skin.

Upon test item request, the One test item was procured from the Test Item Control Office (TICO). For test item application sites, the ready to use test item was cut into size of 6.25 cm^2 (2.5 cm x 2.5 cm) in a Laminar Air Flow Cabinet prior to application and applied onto the clipped surface of the skin.

All the application sites were covered with semi-occlusive dress (6 ply cotton gauze size: 3 cm X 4 cm) and held secured around the torso of the animal by hypoallergenic tape and contact period was 4 hours (+38 minutes). At the end of contact period, dressings were removed and the application sites were cleaned by gently wiping off with cotton moistened with RO water and wiped dry with dry cotton. The positions of the application sites were marked with indelible marker pen. Animals were observed for skin irritation at 1 hour, 24, 48 and 72 hours post-patch removal. Animals were observed for any mortality and morbidity, clinical signs, Erythema and Oedema at pre- and post - patch application, and post-patch removal. Individual animal body weights were measured at the receipt, on Day 1 pre-application and at termination (Day 4).

The application patches and its sites were applied onto each rabbit as shown in the **Section 10.2.1.1**

No mortality, morbidity and clinical signs were observed during the course of the experiment. No skin reactions (Erythema / Oedema) at any of the post-application or post-patch removal observation time points. Therefore, the animals were terminally sacrificed on Day 4. The body weights were unaffected by test item application during the study period.

Based on the Erythema and Oedema evaluation grades across the observation intervals, the Primary Irritation Index (PII) and Cumulative Irritation Index (CII) were both determined to be **0** for the control and test item application sites.

Conclusion: Based on the results, the Primary and Cumulative Irritation Index of test item, "**Drepto Menstroherb Sheet**" determined to be **'0'**. Therefore, test item considered to be **"Non-Irritant"** to the skin of New Zealand White rabbits as per the criteria (i.e. mean score 0 to 0.4 indicative of Negligible) of test method ISO 10993-23:2021/Amd 1:2025 and under the conditions employed.

2. OBJECTIVE

The objective of conducting this *In-Vivo* toxicity study / test was to evaluate the irritancy potential of the Test Item "Drepto Menstroherb Sheet" following single exposure topical application to the intact skin of the rabbit to evaluate the possibility of skin reaction (Erythema and Oedema). This test also provides information on health hazards that were likely to arise from acute exposure by the intended clinical route in humans.

3. STUDY DETAILS

Study Title	:	Skin Irritation Test of Test Item "Drepto Menstroherb Sheet" with New Zealand White Rabbits.
Study Number	:	NLS-TX-270/26
Study Code	:	SIT-RB
ULR No	:	TC-1286626000000431
Sponsor	:	DREPTO BIODEVICES PVT. LTD Off. Sine IIT, Bombay, 1001-T1, Rustomjee Ozone. Co-op HSG, Beh.Inorbit, Goregoan (Mumbai)Mumbai, Goregoan West, Maharashtra India, 400104.
Sponsor Representative & Monitoring Scientist	:	Rahul Kumar Gupta
E- mail	:	info@mcoh.co.in
Test Facility	:	NINKILIM LIFE SCIENCES PRIVATE LIMITED Plot No. 553P-14, Vasanthanarasapura 2nd Phase KIADB Industrial Area, Tumakuru-572128 Karnataka India

4. STUDY RESPONSIBILITIES

Study Director	:	Rohith Kumar H.N.
Study Personnel	:	Govardhana Patel D.L.
Clinical Veterinarian	:	Dr. Ranjith Roy Yadav

5. STUDY SCHEDULE

Study Initiation Date	:	12/08/2026
Experiment Start Date	:	13/08/2026
Acclimatization Period	:	13/08/2026 to 17/08/2026
Treatment Date	:	18/08/2026
Experiment End Date	:	21/08/2026
Draft Report to Sponsor	:	24/08/2026
Study Completion Date	:	02/09/2026

6. TEST METHOD

The design of this test was based on the study objectives and procedures as specified in the Study Plan. The overall product development strategy for the test item, together with the test method described below, was applied in accordance with the relevant principles.

- ISO 10993-23:2021 - Biological Evaluation of Medical Devices-Part 23: Tests for Irritation (First Edition 2021) / Amd 1: 2025.

7. ANIMAL WELFARE AND VETERINARY CARE

Test Facility was registered with CCSEA, Department of Animal Husbandry and Dairying (DAHD), Ministry of Fisheries, Animal Husbandry and Dairying (MoFAH & D), Government of India. Also, Test facility ensures that animal experiments are performed in accordance with the recommendation of the regulatory guidelines / test methods for laboratory animal facility published in the gazette of India, 2023 and 10993-2:2022: Biological evaluation of medical devices-Part 2: Animal welfare requirements.

During the conduct of this test, none of the animals were injured, and no moribund animals were observed.

8. IAEC APPROVAL

The use of animals in this test has been approved by Test facility IAEC.IAEC approved Protocol No.: NIK/IAEC-044/06-2025.

9. AMENDMENT AND DEVIATION PROCEDURES

No amendments or deviations occurred during the conduct of test.

10. MATERIALS AND METHODS

10.1 Materials

10.1.1. Test Item Information

The Test Item Information Data sheet (TIDS) provided by the Sponsor to the Test Facility is as furnished below:

Test Item Name	:	Drepto Menstroherb Sheet
Test Item Manufacturer, Supplier & Batch Produced by	:	Indocoar Pharma Pvt. Ltd RS 56/B/1, Karwar road, Kalaghatgi-Tq, Dharawad Dist-581204. Karnataka, India
Code by Test Facility	:	S67/TI-001
Batch No./ Lot No.	:	REVC2602
Date of Manufacture	:	June-2026
Date of Expiry	:	May-2029

The Identity (TIDS - **Refer to Annexure No. 1**) of the Test Item was provided by the Sponsor Test Facility. The Sponsor is responsible for the authenticity of the test item and Test Facility did not perform any further characterization of the test item.

Note: The remaining test item (if any) from Test Item Control Office (TICO) shall be disposed at the completion of the battery of tests under this Test Item.

10.1.2. Test System

Species	:	Rabbit *(Oryctolagus cuniculus)*
Strain	:	New Zealand White Rabbit
Justification for Selection of Species	:	Rabbit is the recommended species for conducting Skin Irritation Test as per guideline ISO 10993-23:2021/ Amd 1:2025.
Source	:	Vaarunya Biolabs Private Limited No.2B/A,2nd PHASE, KIADB Industrial Area, Big Baniyan Tree Road, Near RR Dental College, Bangaluru, Karnataka, India - 560074.
Age (at treatment)	:	3- 4 months old
Total No. of Animals	:	03 Female (Females were Nulliparous & Non-pregnant.)
Animal Number	:	RBa1045 – RBa1047
Body Weight Range at treatment	:	2.124 - 2.361 kg.

10.1.3. Test System Management

10.1.3.1 Animal Room Preparation

Prior to housing the animals, the experimental room was decontaminated by cleaning, fumigation and microbial load was checked by settle plate method. The experimental room floor was mopped daily once.

10.1.3.2 Husbandry Conditions

Animals were housed in an environment-controlled room temperature of 19.8 - 22.3°C and relative humidity of 41 - 58%, The photoperiod was 12 hours fluorescent light and 12 hours darkness. Adequate fresh air supply of 12 - 15 air changes/hour and sound level of <85 dB was maintained in the experimental room. The relative humidity, maximum and minimum temperature in the experimental room were recorded once daily.

10.1.3.3 Housing

Animals were housed individually in a standard stainless-steel cage (Cage size Length 61 X Breadth 46 X Height 46 cm) provided with stainless-steel wire bottom grill facilitated with polycarbonate resting board, feed hopper and polycarbonate water bottle with stainless-steel sipper tube. Additionally, hanging bells were provided as an enrichment device to minimize the animal stress and promote overall wellbeing of animals. Each litter tray was bedded with steam sterilized corn cob and changed once daily along with tray.

10.1.3.4 Diet and Water

Rabbit Diet manufactured by Krishna Valley Agrotech LLP was provided *ad libitum* to Rabbits. Water subjected to filtration by reverse osmosis was provided *ad libitum* to Rabbits in the polycarbonate bottles with rubber corked stainless-steel sipper tubes.

10.1.4. Test System Preparation

10.1.4.1 Acclimatization

After Veterinary (Physical) examination of animals for good health and those suitable for the study were acclimatized for 5 days prior to treatment. During acclimatization, animals were observed once daily for clinical signs and mortality/morbidity.

10.1.4.2 Animal Identification

During acclimatization period (Temporary identification), each animal was identified by animal number inscribed on the left ear lobe by indelible marker pen. The cages were identified with cage cards indicating study number, study code, species, strain, sex, acclimatization start and acclimatization end date.

During treatment period (Permanent identification), each animal was identified by animal accession number (without prefix) inscribed on the right ear lobe by indelible marker pen. The cages were identified with cage cards indicating study number, study code, species, strain, sex, treatment date, experiment start date and experiment end date.

10.1.4.3 Clipping of Animals

18 hours and 11 minutes of pre-test, the fur on the dorsal region of the animals were closely clipped on both side (right and left side) of the spinal column (10 cm x 15 cm=150 cm^2) by electric hair clipper. During clipping, care was taken to avoid skin abrasions. No skin abrasions observed in any of the animals during the clipping and animals with healthy intact skin were used in the test.

10.1.4.4 Positive Control Response Validation

As per ISO 10993 Part 23:2021/Amd 1:2025, reliability test was conducted with Sodium Lauryl Sulphate (SLS) as a positive control once in 6 Months. The latest positive control response validation study (Reliability Test) results were included in **Annexure No. 4**.

10.2 Methods

10.2.1. Experimental Procedure

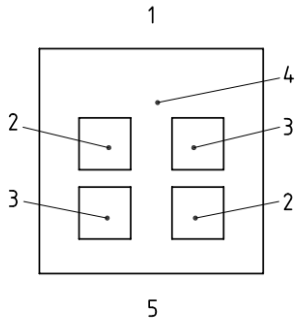
10.2.1.1. Single Exposure Test

The irritation potential of the test item "Drepto Menstroherb Sheet" was evaluated using New Zealand White rabbits. Although the device is intended for surface contact (intact skin) during menstrual use, dermal irritation testing provides a standardized and conservative model for assessing local tolerance. The topical dermal route ensures controlled exposure and allows meaningful evaluation of potential tissue reactions, including Erythema and Oedema. This approach supports the biological safety assessment of the device by generating clinically relevant data on possible local irritant effects associated with patient use.

For control application sites, the filter paper was cut into size of 6.25 cm^2 (2.5 cm x 2.5 cm) in a Laminar Air Flow Cabinet prior to application and was impregnated with RO water and applied onto the clipped surface of the skin.

Upon test item request, the One test item was procured from the Test Item Control Office (TICO). For test item application sites, the ready to use test item cut into size of 6.25 cm^2 (2.5 cm x 2.5 cm) in a Laminar Air Flow Cabinet prior to application and applied onto the clipped surface of the skin.

The application patches and its sites were applied onto each rabbit as depicted below:



Key

1. Cranial End
2. Test Item Site
3. Control Site
4. Clipped Dorsal Region
5. Caudal End

All the application sites were covered with semi-occlusive dress (6 ply cotton gauze size: 3 cm X 4 cm) and held secured around the torso of the animal by hypoallergenic tape and contact period was 4 hours (+38 minutes). At the end of contact period, dressings were removed and the application sites were cleaned by gently wiping off with cotton moistened with RO water and wiped dry with dry cotton. The positions of the application sites were marked with indelible marker pen. Animals were observed for skin irritation at 1 hour, 24, 48 and 72 hours post-removal of applied patch. The unused test item was stored in a container and sent for disposal.

11. OBSERVATIONS

11.1 Observation Period

After removal of acute single exposure patches, the skin reactions were scored at 1 hour, 24, 48 and 72 hours post-patch removal of acute single exposure.

Note: Study was terminated on Day 4 as there were no skin reactions at any of the post-application and post-patch removal observation time points.

11.2 Mortality, Morbidity and Clinical Signs

Animals were observed twice daily for mortality and morbidity, and once daily for clinical signs throughout the observation period. As there were no clinical signs during observation period, animals were observed once daily (morning hours) for the morbidity and the mortality during Weekends and Holidays.

11.3 Body Weight

Individual animal body weights were measured on the day of receipt, on Day 1 (pre-application) and at termination (Day 4).

11.4 Scoring of Skin Reactions

The application sites of each animal were examined for any skin reactions. Scoring of Erythema and Oedema for each application site at each observation interval in accordance with the scoring system provided in **Annexure No 2.**

The detailed skin reactions observations were made at 24, 48 & 72 hours after removal of acute single exposure.

12. SKIN IRRITATION EVALUATION CRITERIA

Single Exposure Primary Irritation Index (PII) was calculated as follows:

- Only (24 ± 2), (48 ± 2) and (72 ± 2) hours observation periods were used for calculations.
- After 72 hours grading, all erythema grades plus oedema grades (24 ± 2), (48 ± 2) and (72 ± 2) hours are summated separately for each test sample and control for each animal.
- The primary irritation score for individual animal was calculated by dividing the sum of all the scores by 6 (2 observation sites x 3 time points).
- Primary Irritation Index for the test item was obtained by summating the individual scores and divided by the number of animals given in **Annexure No. 3.**

13. ANIMAL EUTHANASIA AND DISPOSAL

At the end of experiment period all the animals were sacrificed using lethal dose (50 mg/mL) of Sodium thiopental injection. The carcasses were stored frozen and sent for disposal.

Discerptions	Batch No.	Manufactured Date	Expiry Date	Manufactured By
Sodium Thiopentoneal Injection	N24339A	02/2025	01/2027	American Remedies Healthcare Pvt. Ltd.
Sterile Water for Injection	W2024023	07/2024	08/2028	Shiv Healthcare

14. RESULTS AND DISCUSSION

14.1 Mortality , Morbidity and Clinical Signs

Refer to Table 1 and Appendix 1

No mortality / morbidity till termination. There were no clinical signs observed in any of the animals throughout the experimental period.

14.2 Body Weights

Refer to Table 2 and Appendix 2

A normal increase in body weight (mean and % gain) was observed at termination in all animals, relative to Day 1 pre-application body weight values.

14.3 Skin Reaction Scores

Refer to Table 3 and Appendix 3

No Erythema or Oedema observed in the test item and control application sites in any of the tested animal.

The detailed skin reactions observations were made at 1 hour (+3 minutes), 24 hours (+2 minutes), 48 hours (+6 minutes) and 72 hours (+3 minutes) post-patch removal of acute single exposure.

The mean Erythema and Oedema scores for the test item and control application sites were calculated for each animal at each scoring interval. Based on the Erythema and Oedema evaluation grades across the observation intervals, the Primary Irritation Index (PII) and the Cumulative Irritation Index (CII**)** were both determined to be 0 for the control and test item application sites.

15. CONCLUSION

Based on the results, the Primary and Cumulative Irritation Index of test item, "**Drepto Menstroherb Sheet**" was determined to be **'0'**. Therefore, test item considered to be **"Non-Irritant"** to the skin of New Zealand White rabbits as per the criteria (i.e. mean score 0 to 0.4 indicative of Negligible) of test method ISO 10993-23:2021/Amd 1:2025 and under the conditions employed.

16. DATA COMPILATION

All data are presented in annexures and/or summarized and presented in tables. All findings are presented in the Study Report as per the standard reporting format.

17. STUDY REPORT DISTRIBUTION

The Final Study Report and Test Report will be distributed as follows:

a) Copy No. 1/2 – Test Facility Archives.
b) Copy No. 2/2 –Sponsor.

18. ARCHIVING

All study-related records, Study Plan, Raw Data, Final Study Report and Test Report will be maintained in the archives of Test Facility for 9 years from the date of study completion. The test item sample will be archived till its expiry. All the records and test item will be handled according to OECD Principles of GLP. After the completion of archiving period, the Test Facility will coordinate with the Sponsor for further course of action on archived material.

19. REFERENCES

- ISO/IEC 17025:2017, General Requirements for the Competence of Testing and Calibration Laboratories (Third Edition –Nov 2017).
- ISO 10993-1:2025-Biological Evaluation of Medical Devices-Part-1: Evaluation and testing within a risk management process (Sixth Edition-Nov 2025)
- ISO 10993-2:2022-Biological Evaluation of Medical Devices-Part 2: Animal welfare requirements (Third Edition 2022-11).
- ISO 10993-23:2021: Biological evaluation of medical devices-Part 23: Tests for Irritation (First Edition 2021-01) / Amd 1: 2025.
- OECD Principles of GLP for testing of chemicals as specified by international [(C (97) 186/Final] legislation.
- ISO/IEC 17025:2017, General Requirements for the Competence of Testing and Calibration Laboratories.
- The mutually agreed Study Plan and the Standard Operating Procedures (SOPs) of Test Facility.

Table 1. Summary of Clinical signs

Refer to Appendix 1

Animal No.	Sex	Clinical Signs observations (Day 1-4)	Mortality & Morbidity
RBa1045	Female	N	0/1
RBa1046		N	0/1
RBa1047		N	0/1

Key: N-Normal, 0'-No mortality observed out of '1' animals

Table 2. Summary of Body Weight gain (%)

Refer to Appendix 2

Animal No.	Sex	Body Weight Gain (%)
		Days (1-4)
RBa1045	Female	0.79
RBa1046		0.97
RBa1047		0.89
Mean		**0.89**
SD		**0.09**

Table 3. Summary of Skin Reaction

Refer to Appendix 3

Animal No.	Sex	Test Item Sites				Control Sites				PIS
		1 hour	24 hours	48 hours	72 hours	1 hour	24 hours	48 hours	72 hours	
RBa1045	Female	0	0	0	0	0	0	0	0	0
RBa1046		0	0	0	0	0	0	0	0	0
RBa1047		0	0	0	0	0	0	0	0	0

Key: PIS-Primary irritation score

Appendix 1. Individual Animal Clinical Signs

Animal No.	Sex	No. of Mortality / Morbidity				Clinical Signs			
		Day 1	Day 2	Day 3	Day 4	Day 1	Day 2	Day 3	Day 4
RBa1045	Female	0	0	0	0	N	N	N	N
RBa1046		0	0	0	0	N	N	N	N
RBa1047		0	0	0	0	N	N	N	N

Key: N-Normal, 0-No Mortality/Morbidity

Note: All animals were found normal during the course the Acclimatization period.

Appendix 2. Individual Animal Body Weight

Animal No.	Sex	Body Weight (kg)	
		Day 1	Day 4
RBa1045	Female	2.283	2.301
RBa1046		2.361	2.384
RBa1047		2.124	2.143
Mean		**2.26**	**2.28**
SD		**0.12**	**0.12**

Appendix 3. Evaluation of Skin Reaction

Animal No. and Sex	RBa1045 and Female								RBa1046 and Female								RBa1047 and Female							
Application Site	Test Sites				Control Sites				Test Sites				Control Sites				Test Sites				Control Sites			
	2(a)		2(b)		3(a)		3(b)		2(a)		2(b)		3(a)		3(b)		2(a)		2(b)		3(a)		3(b)	
Observation	E	O	E	O	E	O	E	O	E	O	E	O	E	O	E	O	E	O	E	O	E	O	E	O
Day 1 (1 hour)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Day 2 (24 hours)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Day 3 (48 hours)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Day 4 (72 hours)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Erythema + Oedema	0				0				0				0				0				0			
Primary Irritation Score of Individual Animal (Test Site - Control Site)	0								0								0							
Primary Irritation Index	0.00																							

Key: 2: test site; 3: control sites, (a): Left flank & 2 (b): Right flank, E: Erythema; O: Oedema

$$\text{Primary Irritation Score} = \frac{\text{Total of Erythema and Oedema Grades at } (24 \pm 2), (48 \pm 2) \text{ and } (72 \pm 2) \text{ hours for Test Item and Control Sites}}{6 \text{ (2 Observation Sites x 3 Time Points)}}$$

$$\text{Primary Irritation Index} = \frac{\text{Sum of Primary Irritation Scores of all Animals}}{3 \text{ Animals}}$$

Annexure No. 1. Test Item Information Data Sheet (TIDS)

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TEST ITEM INFORMATION DATA SHEET(TIDS)

1.	Sponsor (Name and Address to be used in the report)	Drepto Biodevices Pvt. Ltd Off. Sine IIT, Bombay, 1001-T1, Rustomjee Ozone. Co-op HSG, Beh.Inorbit, Goregoan (Mumbai), Mumbai, Goregoan West, Maharashtra, India,400104.
2.	Name of Test Item (to be included in the report)	Drepto Mentroherb Sheet
3.	Category of Test Substances: Medical Device	
4.	Manufacturer (Name and Address, if different than 1)	Indocoar Pharma Pvt. Ltd, RS 56/B/1, Karwar road, Kalaghatgi-Tq, Dharawad Dist-581204. Karnataka, India
5.	Supplier	Same as serial No.4
6.	Address for Communication with E-mail:	Indocoar Pharma Pvt. Ltd, N. 10, Amarjyothi layout, Ist Main Road, Geddalahalli, Sanjaynagar, Bangalore-560094. Tel No.+91 8023512750 Email: info@mcoh.co.in
7.	Batch No./ Lot No.	REVC2602
8.	Date of Manufacture / Batch produced on	JUN-2026
9.	Batch produced by (Name and address)	Same as serial No.4
10.	Date of Expiry	MAY-2029
11.	Date of Retest	NA
12.	Chemical Name (IUPAC)	NA
13.	CAS No.	NA
14.	Physical appearance (with colour)	A pale yellow adhesive which is consisting of drug and covered with fabric and release liner
15.	Purity (Declared by sponsor and/or as per Certificate of Analysis/ assay of Active Ingredient (AI) if applicable)	Boswellia serrata-108.30 % Eucalyptus oil -100.18 % Menthol-100.0% Curcumin -100.13%
16.	Recommended Storage conditions	Ambient (25 ± 4 °C)
17.	p^{H} (if applicable)	Not applicable
18.	Density / Specific gravity (if applicable)	Not applicable
19.	Type of material: Transdermal patch (Specify)	
20.	Type of package: Triple laminated Aluminum pouch	
21.	Intended Clinical Use (as in the device label/IFU-information for use)	The Drepto Mentroherb sheet is Providing relief from pain by absorbing and Transderma contact with stomach surface.



Annexure No 1. (Cont'd) Test Item Information Data Sheet (TIDS)

Ninkilim Life Sciences Pvt Ltd

22.	Test Item Categorization: 1. Surface: Intact skin 2. Contract Duration: <24 h	
23.	Test Item form	Finished form
24.	Test Item dimension (fill appropriate)	Surface area (in mm^2 /cm^2): 75 cm^2 Length (in mm/cm):15.0 cm Breadth (in mm/cm):5.0 Weight (in mg/g): About : 2.7403 g
25.	Whether the test item can be cut as needed for testing	☑ Yes ☐ No
26.	Extraction ratio to be used for testing	Not applicable
27.	Extraction condition to be used for testing	Not applicable
28.	Test Item Sterilization method adopted	Non-sterile
29.	If non-sterile, select the method of sterilization	☐ Autoclave Method ☐ Surface Sterilization ☑Tested on Non-sterile
30.	List of part(s) of the test item that should be tested (if applicable)	Not applicable
31.	A list of part(s) of the test item should be excluded from the test (if applicable)	Not applicable
Others		
32.	Particulars of dispatch: 1.Quantity and Date 2. Mode of Dispatch	10 Pouch X 2 (20 No's) 01/08/2026 Courier
33.	Certificate of Analysis / Analytical Test Report attached	☐ Yes ☐ No ☑NA
34.	Information for use is attached	☐ Yes ☐ No ☑NA
35.	The Material Safety Data Sheet (MSDS)is attached	☐ Yes ☐ No ☑NA
36.	Hazards and precautions (if any other than those specified in MSDS or if MSDS is not provided)	Not applicable
37.	Consent to disposal of leftover test item Consent to disposal of Archived test item on expiry	Yes Yes
38.	Any additional information (optional)	Not applicable
39.	Name of the Sponsor's Representative	Rahul Kumar Gupta
40.	Name of the Monitoring Scientist	Rahul Kumar Gupta & Dr. Chandan Yadav

Annexure No 1. (Cont'd) Test Item Information Data Sheet (TIDS)

Ninkilim Life Sciences Pvt Ltd

List out the test to be conducted:		
Sl. No.	**Test Title**	**Test Method**
1.	Invitro Cytotoxicity Test (DCM)	ISO 10993 Part 5:2009
2.	Skin Irritation Test (SIT-RB)	ISO 10993 Part 23:2021/Amd1:2025
3.	Skin Sensitization (BT)	ISO 10993-10:2021

Asa SponsororSponsorrepresentative ofthesestudies, I agreewithbelowpoints:

- Thestudiesrequestedare tomeet theregulatoryrequirementsofthe testitem.
- The animal usage is necessary for requested studies as per guideline requirements. The species chosen is appropriate to the study and as per guidelines requirements.
- The studies requested are not an unnecessary duplication of previous work.

SponsororSponsorRepresentative: Rahul Kumar Gupta

DREPTO BIODEVICES PVT LTD MUMBAI 400 076

Sign. and Date 07/08/25

InstructionsforfillingTestItem/ReferenceItem/Vehicle/ Placebo DataSheet:

- Fillthe informationsheetwithavailableinformation.
- Iftheinformationisnotavailablementionit asNAandifthe sectionorcolumnisnotapplicableforthe testitem,referenceitem,orvehicle,mentionit asNA(NotApplicable).

Addcolumnsor rows asperrequire

Annexure No 2. Scoring System for Skin Reaction

Reaction	Irritation Score
Erythema and Eschar Formation	
No erythema	0
Very slight erythema (Barely perceptible)	1
Well-defined erythema	2
Moderate erythema	3
Severe erythema (beet-redness) to eschar formation preventing grading of erythema	4
Oedema Formation	
No oedema	0
Very slight oedema (Barely perceptible)	1
Well defined oedema (edges of area well defined by definite raising)	2
Moderate oedema (raised approximately 1 mm)	3
Severe oedema (raised more than 1 mm and extending beyond exposure area)	4
Maximal Possible Score for Irritation	**8**
Other adverse changes at the skin sites shall be recorded and reported	

Annexure No 3. Primary or Cumulative Irritation Index Categories in a Rabbit

Primary Irritation Index (PII)	
Mean Score	**Response Category**
0 to 0.4	Negligible
0.5 to 1.9	Slight
2 to 4.9	Moderate
5 to 8	Severe

Annexure No 4. Positive Control Response Validation

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Annexure No 5. Summary Results of Positive Control (SLS) Study

Study Number	:	NIK-TX-1008
Positive Control Name (Test Item)	:	Sodium Lauryl Sulphate (SLS)
Batch No.	:	PF/23/2014/119-13/24/Q400/16
Experimental Period	:	23/04/2026 to 12/05/2026
Valid Up to	:	12/11/2026

The objective of conducting this *In-Vivo* toxicity test / study was to evaluate the Irritancy Potential and Reliability of the test procedure with known Positive Control (Test Item) "Sodium Lauryl Sulphate (SLS)" followed by single exposure of topical application on to the rabbit intact skin to evaluate the possibility of skin reaction (i.e., Erythema and Oedema) with New Zealand White Rabbits This test was conducted in accordance with Test Method ISO 10993-23:2021/Amd 1:2025- Biological Evaluation of Medical Devices – Test for Irritation.

Four Female rabbits were acclimatized for 5 Days. Three Female rabbits were selected for application. Prior to the 19 hours and 49 minutes of application, the fur of the spine (dorsal flank region) of the animals were closely clipped (i.e., 10 cm x 15 cm) on both sides (right and left) of the spinal columns by electric hair clipper. Care was taken to prevent any skin abrasion, and no skin abrasion were observed on any of the animal.

For all three rabbits were applied onto the clipped area skin of both sides. For control application sites, 0.5 mL of distilled water was impregnated onto the cotton gauze (6 cm^2 i.e. 2 x 3 cm of 6 ply) and applied onto the clipped area of desginagated skin sites. For test item (SLS) sites, the 0.5 g of SLS powder was weighed onto an aluminum foil and made into paste by adding the 0.5 mL of distilled water with the help of glass rod. The prepared paste was completely transferred onto the cotton gauze (6 cm^2 i.e. 2 x 3 cm of 6 ply) and and applied onto the desginagated sites. All the application patches were covered with the help of semi-occlusive dressings and held secured around the torso of the animal by hypoallergenic tape and contact period was 4 hours (+ 4 minutes). At the end of contact period, dressed patch was removed. The application sites were cleaned by gently wiped off with cotton moistened with RO water and pat dry with dry cotton. The application sites were marked with indelible marker pen for easy of observations.

All the animals were observed for mortality and morbidity and clinical sings. The animals were observed for skin irritation (Erythema and Oedema) at 1hour, 24 hours, 48 hours, 72 hours post-patch removal of scheduled observations and till termination. Individual animal body weight was measured at receipt, on Day 1 pre-application and at termination (Day 15).

No mortality/morbidity and clinical signs were observed during the course of the experiment. The body weight measured during the course of the test was unaffected by the test item application. No Erythema and/or Oedema was observed in vehicle control treated sites till termination (Day 15) whereas in Positive control (SLS) sites, very slight to severe Erythema and very slight Oedema (Barely perceptible) to well-defined oedema (edges of area well defined

Annexure No 4. (Contd.,) Positive Control Response Validation

Ninkilim Life Sciences Pvt Ltd

by definite raising) was observed in all animals (3/3) at 1, 24, 48 & 72 hours (i.e., till Day 4 post-application), moderate to severe erythema was continued to be observed up to Day 8 of post-application and well-defined Erythema was observed on Day 9 & 10. The very slight Erythema was observed from Day 11 to till terminal Day 15. However, the observed reaction/s were Indurated (hardened) on Day 5 (i.e., at 96 hours).

The primary Irritation Score of the vehicle control was "0" (3/3) and test item score for animal No. RBa756, RBa757 and RBa758 was 4.5, 4.8 and 4.7, respectively.

The Primary Irritation Index (PII) of test item (Positive control - SLS) sites was 4.5+4.8+4.7 / 3= 4.7. As per PII criteria, the mean score of 2 to 4.9 is considered as response category of Moderate Irritation.

Conclusion: Based on the above results, the obtained Positive control, **"Sodium Lauryl Sulphate"** Primary Irritation Index was 4.7.Therefore, considered as **"Moderate Irritant"** under the test conditions employed. This confirms both the irritancy potential of SLS and the reliability of the test procedure conducted in accordance with ISO 10993-23:2021/Amd 1:2025

13/05/2026

Govardhana Patel D.L.
Study Director
Sign. and Date

Annexure No 5. Study Plan

Ninkilim
Life Sciences Pvt Ltd

STUDY PLAN

STUDY TITLE

SKIN IRRITATION TEST OF TEST ITEM "DREPTO MENSTROHERB SHEET" WITH NEW ZEALAND WHITE RABBITS

TEST METHOD: ISO 10993-23:2021 / Amd 1:2025

STUDY No.: NLS-TX-270/26

STUDY CODE: SIT-RB

STUDY DIRECTOR

Rohith Kumar H.N.

SPONSOR

DREPTO BIODEVICES PVT. LTD
OFF. SINE IIT, BOMBAY, 1001-T1, RUSTOMJEE OZONE
CO-OP HSG, BEH.INORBIT, GOREGOAN (MUMBAI)
MUMBAI, GOREGOAN WEST, MAHARASHTRA
INDIA, 400104.

TEST FACILITY

NINKILIM LIFE SCIENCES PRIVATE LIMITED
PLOT No. 553P-14, VASANTHANARSAPURA
2nd PHASE, KIADB INDUSTRIAL AREA
TUMAKURU-572128
KARNATAKA
INDIA

Annexure No 5. (Contd.,) Study Plan

Ninkilim Life Sciences Pvt Ltd

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Annexure No 5. (Contd.,) Study Plan

Ninkilim Life Sciences Pvt Ltd

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Annexure No 5. (Contd.,) Study Plan

Ninkilim
Life Sciences Pvt Ltd

1. OBJECTIVE

The objective of conducting this *In-Vivo* toxicity study / test is to evaluate the irritancy potential of the Test Item "Drepto Menstroherb Sheet" following single exposure topical application to the intact skin of the rabbit to evaluate the possibility of skin reaction (Erythema and Oedema). This test also provides information on health hazards that are likely to arise from acute exposure by the intended clinical route in humans.

2. STUDY DETAILS

Study Title	:	Skin Irritation Test of Test Item "Drepto Menstroherb Sheet" with New Zealand White Rabbits.
Study Number	:	NLS-TX-270/26
Study Code	:	SIT-RB
ULR No	:	TC-1286626000000431
Sponsor	:	DREPTO BIODEVICES PVT. LTD Off. Sine IIT, Bombay, 1001-T1, Rustomjee Ozone. Co-op HSG, Beh.Inorbit, Goregoan (Mumbai)Mumbai, Goregoan West, Maharashtra India, 400104.
Sponsor Representative & Monitoring Scientist	:	Rahul Kumar Gupta
E- mail	:	info@mcoh.co.in
Test Facility	:	NINKILIM LIFE SCIENCES PRIVATE LIMITED Plot No. 553P-14, Vasanthanarasapura 2nd Phase KIADB Industrial Area, Tumakuru-572128 Karnataka India

Annexure No 5. (Contd.,) Study Plan

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Life Sciences Pvt Ltd

3. STUDY RESPONSIBILITIES

Study Director	:	Rohith Kumar H.N.
Clinical Veterinarian	:	Dr. Ranjith Roy Yadav

4. STUDY SCHEDULE

Study Initiation Date	:	12/08/2026
Experiment Start Date	:	13/08/2026
Acclimatization Period	:	13/08/2026 to 17/08/2026
Treatment Date	:	18/08/2026
Experiment End Date	:	Latest by 01/09/2026
Draft Report to Sponsor	:	Latest by 04/09/2026
Study Completion Date	:	Latest by 10/09/2026 or within a Week after receipt of comments on Draft Report from Sponsor or on the date of Study Director signature.

Annexure No 5. (Contd.,) Study Plan

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5. ABBREVIATIONS AND SYMBOLS

Amd	:	Amendment
CCSEA	:	Committee for Control and Supervision of Experiments on Animals
cm	:	Centimetre
dB	:	Decibel
GLP	:	Good Laboratory Practice
gm	:	gram
IAEC	:	Institutional Animal Ethics Committee
IEC	:	International Electrotechnical Commission
SIT-RB	:	Skin Irritation Test with New Zealand White Rabbits
ISO	:	International Organization for Standardization
NLS	:	Ninkilim Life Sciences Private Limited
mg	:	Milligram
mL	:	Millilitre
mm	:	Millimetre
OECD	:	Organization for Economic Co-operation and Development
RO	:	Reverse Osmosis
SOP	:	Standard operating procedure
Sign	:	Signature
TIDS	:	Test Item Information Data Sheet
TX	:	Toxicology
ULR	:	Unique Lab Report
±	:	Plus or Minus
&	:	and
≤	:	Less than or equal to
%	:	Percentage
°C	:	Degree Celsius

Note: Additional abbreviations and symbols (if any) will be provided in Study Report.

6. QUALITY ASSURANCE UNIT RESPONSIBILITIES

The Quality Assurance Unit (QAU) of the Test Facility will inspect different phases of the study to review the Study Plans, Amendments (if any), raw data, Study Reports to assure the integrity of study in compliance with ISO/IEC 17025:2017 and OECD Principles of GLP. Findings of all the inspections during the study will be recorded and reported to the Study Director (SD) and Test Facility Management (TFM). A statement of Quality Assurance Unit will be provided in the Study Report.

7. STUDY COMPLIANCE

The test will be performed in compliance with the following:

- ISO/IEC 17025:2017, General Requirements for the Competence of Testing and Calibration Laboratories.
- OECD Principles of GLP for testing of chemicals as specified by international [(C (97) 186/Final] legislation.
- The mutually agreed Study Plan and the Standard Operating Procedures (SOPs) of Test Facility.

8. TEST METHOD

The design of this test is based on the study objectives and procedures as specified in the Study Plan. The overall product development strategy for the test item, together with the test method described below, is applied in accordance with the relevant principles.

- ISO 10993-23:2021: Biological evaluation of medical devices-Part 23: Tests for Irritation (First Edition 2021-01) / Amd 1: 2025.

9. ANIMAL WELFARE AND VETERINARY CARE

Test Facility is registered with CCSEA, Department of Animal Husbandry and Dairying (DAHD), Ministry of Fisheries, Animal Husbandry and Dairying (MoFAH & D), Government of India. Also, Test Facility ensures that animal experiments are performed in accordance with the recommendation of the regulatory guidelines/ test methods for laboratory animal facility published in the Gazette of India, 2023 and 10993-2:2022: Biological evaluation of medical devices-Part 2: Animal welfare requirements.

During this test if any animal gets injured, ill or moribund, care will be taken as per the current veterinary practices. If required, for humane reasons, animals will be euthanized as per the standard procedures. The objective of the study will be considered before any decision and event of any unlikely situations will be intimated to the sponsor.

10. IAEC APPROVAL

The use of animals in this test has been approved by Test Facility IAEC. IAEC approved Protocol No.: NIK/IAEC-044/06-2025. Any significant changes to this Study Plan will be intimated to IAEC and approvals will be sought subsequently, if required.

Annexure No 5. (Contd.,) Study Plan

Ninkilim Life Sciences Pvt Ltd

11. AMENDMENT AND DEVIATION PROCEDURES

This Study Plan may be amended or subjected to alterations.

The amendment to the approved Study Plan will be put in writing and realized only after written consent from the Study Sponsor. If immediate action is necessary, verbal agreement with the Sponsor will be confirmed as soon as possible by Study Plan Amendment.

Any unintended change(s) to the Study Plan will be documented in the raw data and mentioned in the report as deviation(s).

12. SAFETY PRECAUTIONS

The personnel involved in study conduct will be wearing all necessary personnel protective equipment such as gloves, head cap, goggles (if required) and face mask in addition to protective body garments and slippers/shoes to ensure adequate personnel health and safety and to avoid inhalation and skin contact with the test item.

13. MATERIALS AND METHODS

13.1 Materials

13.1.1 Test Item Information

The Test Item Information Data Sheet (TIDS) provided by the Sponsor to the Test Facility is as furnished below:

Test Item Name	:	Drepto Menstroherb Sheet
Test Item Manufacturer, Supplier & Batch Produced by	:	Indocoar Pharma Pvt. Ltd RS 56/B/1, Karwar road, Kalaghatgi-Tq, Dharawad Dist-581204. Karnataka, India
Code by Test Facility	:	S67/TI-001
Batch No./ Lot No.	:	REVC2602
Date of Manufacture	:	June-2026
Date of Expiry	:	May-2029

The identity (TIDS - Refer to **Annexure No. 1**) of the Test Item is provided by the Sponsor Test Facility. The Sponsor is responsible for the authenticity of the test item and Test Facility will not perform further characterization of the test item.

Note: The remaining test item (if any) from Test Item Control Office (TICO) will be disposed at the completion of the battery of test under this Test Item.

Annexure No 5. (Contd.,) Study Plan

Ninkilim Life Sciences Pvt Ltd

13.1.2 Test System

Species	:	Rabbit *(Oryctolagus Cuniculus)*
Strain	:	New Zealand White
Justification for Selection of Species	:	Rabbit is one of the recommended species for conducting intracutaneous reactivity test as per Test Method ISO-10993-23:2021 / Amd 1:2025
Source	:	Vaarunya Biolabs Private Limited No.2B/A,2nd PHASE, KIADB Industrial Area, Big Baniyan Tree Road, Near RR Dental College, Bangaluru, Karnataka, India - 560074.
Age (at treatment)	:	About 3 - 4 months (exact age will be provided in the Study Report)
Total Number of Animals	:	03 Females Females will be Nulliparous & Non-pregnant. Female animals are selected for this study, as the intended clinical use of the device is for Women, and the test system must reflect the target population.
Animal Number	:	RBa1045 – RBa1047
Body Weight range (at treatment)	:	Not Less than 2 kg (exact weight will be provided in the Study Report)

13.1.3 Test System Management

13.1.3.1 Animal Room Preparation

Prior to housing the animals, the experimental room will be decontaminated by cleaning, fumigation and microbial load will be checked by settle plate method. The experimental room floor will be mopped once daily.

13.1.3.2 Husbandry Conditions

Animals will be housed in an environment-controlled room temperature of 20 ± 3 °C and relative humidity of 30-70%. The photoperiod will be 12 hours fluorescent light and 12 hours darkness. Adequate fresh air supply of 12 - 15 air changes/hour and sound level of ≤85 dB will be maintained in the experimental room.

The relative humidity, maximum and minimum temperature in the experimental room will be recorded once daily.

13.1.3.3 Housing

Animals will be housed individually in a standard stainless-steel cage (Approximate cage size: Length 61 x Breadth 46 x Height 46 cm) provided with stainless-steel wire bottom grill facilitated to hold the polycarbonate resting board, feed hopper and polycarbonate water bottle corked with stainless-steel sipper tube and stainless-steel litter trays placed below the cage. Each litter tray bedded with steam sterilized corn cob and changed daily along with tray. Additionally, hanging bells will be provided as an enrichment device to minimize animal stress and promote overall wellbeing of animals.

13.1.3.4 Diet and Water

Rabbit Diets manufactured by Krishna Valley Agrotech LLP will be provided *ad libitum* to Rabbit or others will be specified in the Study Report. Water subjected to reverse osmosis will be provided *ad libitum* to Rabbits in polycarbonate bottles rubber corked with stainless-steel sipper tube.

13.1.4 Test System Preparation

13.1.4.1 Acclimatization

After examination for good health and the suitability for the study, the animal will be acclimatized at least for 5 days prior to treatment. During acclimatization, animals will be observed clinical signs, mortality / morbidity at least once daily. Veterinary (Physical) examination will be performed before selecting the animals for the study.

13.1.4.2 Animal Identification

During acclimatization period (Temporary identification), each animal will be identified by ear marking with animal number inscribed on the left ear lobe by Indelible marker pen. The cages will be identified with cage cards at least indicating study number, study code, species, strain, sex, acclimatization start and acclimatization end date.

During treatment period (Permanent identification), each animal will be identified by ear marking with animal accession number (without prefix) inscribed on the right ear lobe by Indelible marker pen. The cages will be identified with cage cards at least indicating study number, study code, species, strain, sex, treatment date, experiment start date and experiment end date.

13.1.4.3 Clipping of Animals

Animals with healthy intact skin will be selected and within 24 to 4 hours prior to test, fur both sides of the spine (flank region) of the animals will be removed (10 cm x 15 cm = 150 cm^2) by clipping closely using electric hair clipper. Care will be taken to avoid abrasion of the skin.

13.1.4.4 Positive Control Response Validation

As per ISO 10993 Part 23:2021/Amd 1:2025, reliability test was conducted with Sodium Lauryl Sulphate (SLS) as a positive control once in 6 Months. The latest positive control response validation study (Reliability Test) results will be presented in the Study Report.

13.2 **Methods**

13.2.1 **Experimental Procedures**

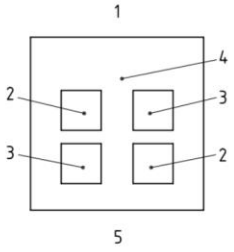
13.2.1.1 **Single Exposure Test**

The irritation potential of the test item "Drepto Menstroherb Sheet" will be evaluated using New Zealand White rabbits. Although the device is intended for surface contact (intact skin) during menstrual use, dermal irritation testing provides a standardized and conservative model for assessing local tolerance. The topical dermal route ensures controlled exposure and allows meaningful evaluation of potential tissue reactions, including Erythema and Oedema. This approach supports the biological safety assessment of the device by generating clinically relevant data on possible local irritant effects associated with patient use.

For control application sites, the filter paper cut into size of 2.5 cm x 2.5 cm (6.25 cm^2) and then impregnated with 0.5 mL of RO water and will be applied onto the skin.

Upon test item request, the required numbers of test item will be procured from the Test Item Control Office (TICO). For test item application sites, the supplied ready to use test item will be cut into size of 2.5 cm x 2.5 cm (6.25 cm^2) and will be applied onto the skin.

The application patches and its sites will be applied on to each rabbit as depicted below:



Key

1. Cranial End
2. Test Item Site
3. Control Site
4. Clipped Dorsal Region
5. Caudal End

All the application sties will be covered with cotton gauze (size: 3 cm x 4 cm of 6 ply) with the help of semi-occlusive dressings and held secured around the torso of the animal by hypoallergenic tape. The contact period will be minimum of 4 hours. At the end of contact period, dressings will be removed. The application sites will be cleaned by gently wiping off with cotton moistened with RO water and pat with dry cotton and return to the respective home-cage. The positions of the application sites will be marked with indelible marker pen. Animals will be observed for skin irritation at (1 ± 0,1) hour, (24 ± 2), (48 ± 2) and (72 ± 2) hours post-patch removal. The unused test item will be stored in a container and send for disposal.

Annexure No 5. (Contd.,) Study Plan

Ninkilim Life Sciences Pvt Ltd

14. OBSERVATIONS

14.1 Mortality, Morbidity and Clinical Signs

Animals will be observed twice daily for mortality, morbidity and at least once daily for clinical signs throughout the observation period. Based on the assessment, if there are no toxic signs / skin reactions of concern, the observation for mortality and morbidity will be once during Weekends and/or Holidays.

14.2 Body Weight

Individual animal body weight will be measured on the day of receipt, on Day 1 pre-application and at termination on Day 4 or as per the requirements recorded in the raw data.

14.3 Skin Reactions

The application sites of each animal will be observed carefully for visualizing the skin reactions. Scoring of skin reactions will be done for Erythema and Oedema for each application site at each time interval as per the scoring system given in the **Annexure No. 2**

The detailed skin reactions observations will be made at about 1 hour, 24, 48 & 72 hours after removal of acute single exposure.

Note: If no skin reactions observed at any of the post-application observation time points, the study will be terminated on Day 4. If any persistent skin reactions are observed, the observation period will be extended up to period of 14 Days.

15. EVALUATION CRITERIA

Single Exposure Primary Irritation Index (PII) will be calculated as follows:

- Only (24 ± 2), (48 ± 2) and (72 ± 2) hours observation periods will be used for calculations.
- After 72 hours grading, all Erythema grades plus Oedema grades (24 ± 2), (48 ± 2) and (72 ± 2) hours will be summated separately for each test item and control for each animal.
- The primary irritation score for individual animal will be calculated by dividing the sum of all the scores by 6 (2 observation sites x 3 time points).
- Primary Irritation Index (Refer to **Annexure No. 3**) for the test item will be obtained by summating the individual scores divided by the total number of animals.

16. ANIMAL EUTHANASIA AND DISPOSAL

At the end of experiment period all the animals will be sacrificed using lethal dose (50 mg/mL) of Sodium thiopental injection. The carcasses will be stored in deep frozen until disposal. The details of sodium thiopental injection and their diluent (SWFI) will be documented in raw data and presented in the Study Report.

Annexure No 5. (Contd.,) Study Plan

Ninkilim Life Sciences Pvt Ltd

17. DATA COMPILATION

All individual animal data will be presented in appendices and / or summarized and presented in tables. All findings will be presented in the Study Report as per the standard reporting format.

18. STUDY REPORT

The Final Study Report will include but is not limited to the following information, as appropriate:

- The descriptive title.
- The name and address of the Sponsor and the Test Facility along with the study schedule.
- List of Scientists/Professionals.
- Statement of study compliance and confidentiality by Study Director.
- The Quality Assurance Statement signed by Quality Assurance Unit.
- Statement of Test Facility Management.
- The test item and its code.
- A description of the test system, including species, strain, source, number, sex, age at start of the treatment, body weight range, housing conditions, and method of identification.
- Positive control results
- Test conditions:
  - Details of test item information
- Observation and Results:
  - Tabulation of response data and dose level for each animal (i.e., tables of clinical signs and mortality, rabbit showing signs of toxicity (if any) including Skin reactions, mortality, nature, severity, and duration of effects)
- Individual Body weights of rabbit and % gain.
- Discussion and conclusion.
- A description of all circumstances if any, that may have affected the quality or integrity of the study.
- The storage location of all raw data, the report, a sample of test item and the archiving period.
- Copy of the final signed Study Plan
- Study Plan amendments if any.
- Study Plan deviations, if any.

Note: If response from Sponsor for finalization of Draft Study Report is not received within Six months after sending draft report in spite of repeated reminders, Test Facility Management could approve termination of study, or finalization of Study Report and archiving. In such an event, any subsequent requests from Sponsor for modification, correction or addition to the Final Study Report will be subject to Study Report amendment at additional cost.

Annexure No 5. (Contd.,) Study Plan

Ninkilim Life Sciences Pvt Ltd

19. STUDY PLAN DISTRIBUTION

The Final Study Plan will be distributed as follows:

a) Copy No.1/2 – Test Facility Archives.
b) Copy No. 2/2 – Sponsor.

20. STUDY REPORT DISTRIBUTION

The Final Study Report and Test Report will be distributed as follows:

a) Copy No.1/2 – Test Facility Archives.
b) Copy No. 2/2 – Sponsor.

21. ARCHIVING

All study-related records, Study Plan, Study Plan amendment and Deviation (if any), Raw Data, Final Report & Test Report, will be maintained in the archives of Test Facility for 9 years from the date of study completion. The test item sample will be archived till its expiry. All the records and test item will be handled according to OECD Principles of GLP. After the completion of archiving period, the Test Facility will coordinate with the Sponsor for further course of action on archived material.

22. REFERENCES

- ISO/IEC 17025:2017, General Requirements for the Competence of Testing and Calibration Laboratories (Third Edition –Nov 2017).
- ISO 10993-1:2025-Biological Evaluation of Medical Devices-Part-1: Evaluation and testing within a risk management process (Sixth Edition-Nov 2025)
- ISO 10993-2:2022-Biological Evaluation of Medical Devices-Part 2: Animal welfare requirements (Third Edition 2022-11).
- ISO 10993-23:2021: Biological evaluation of medical devices-Part 23: Tests for Irritation (First Edition 2021-01) / Amd 1: 2025.
- OECD Principles of GLP for testing of chemicals as specified by international [(C (97) 186/Final] legislation.
- ISO/IEC 17025:2017, General Requirements for the Competence of Testing and Calibration Laboratories.
- The mutually agreed Study Plan and the Standard Operating Procedures (SOPs) of Test Facility.

Annexure No 5. (Contd.,) Study Plan

Ninkilim Life Sciences Pvt Ltd

23. STUDY PLAN APPROVAL

The Study Plan for the Study No.: 'NLS-TX-270/26 has been mutually agreed by the Sponsor through e-mail on 11/08/2026 and approved by the Study Director:

Rohith Kumar H.N.
Study Director

12/08/2026
(Sign. & Date)

Keerthi Kumar R.H.
Quality Assurance Unit

Keerthi Kumar RH 12/08/2026
(Sign. & Date)

Jyothi G.
Test Facility Management

12/08/2026
(Sign. & Date)

Annexure No 5. (Contd.,) Study Plan

Ninkilim Life Sciences Pvt Ltd

Annexure No. 1. Test Item Information Data Sheet (TIDS)

Ninkilim Life Sciences Pvt Ltd

TEST ITEM INFORMATION DATA SHEET(TIDS)

1.	Sponsor (Name and Address to be used in the report)	Drepto Biodevices Pvt. Ltd Off. Sine IIT, Bombay, 1001-T1, Rustomjee Ozone. Co-op HSG, Beh.Inorbit, Goregoan (Mumbai), Mumbai, Goregoan West, Maharashtra, India,400104.
2.	Name of Test Item (to be included in the report)	Drepto Mentroherb Sheet
3.	Category of Test Substances: Medical Device	
4.	Manufacturer (Name and Address, if different than 1)	Indocoar Pharma Pvt. Ltd, RS 56/B/1, Karwar road, Kalaghatgi-Tq, Dharawad Dist-581204. Karnataka, India
5.	Supplier	Same as serial No.4
6.	Address for Communication with E-mail:	Indocoar Pharma Pvt. Ltd, N. 10, Amarjyothi layout, Ist Main Road, Geddalahalli, Sanjaynagar, Bangalore-560094. Tel No.+91 8023512750 Email: info@mcoh.co.in
7.	Batch No./ Lot No.	REVC2602
8.	Date of Manufacture / Batch produced on	JUN-2026
9.	Batch produced by (Name and address)	Same as serial No.4
10.	Date of Expiry	MAY-2029
11.	Date of Retest	NA
12.	Chemical Name (IUPAC)	NA
13.	CAS No.	NA
14.	Physical appearance (with colour)	A pale yellow adhesive which is consisting of drug and covered with fabric and release liner
15.	Purity (Declared by sponsor and/or as per Certificate of Analysis/ assay of Active Ingredient (AI) if applicable)	Boswellia serrata-108.30 % Eucalyptus oil -100.18 % Menthol-100.0% Curcumin -100.13%
16.	Recommended Storage conditions	Ambient (25 ± 4 °C)
17.	p^H (if applicable)	Not applicable
18.	Density / Specific gravity (if applicable)	Not applicable
19.	Type of material: Transdermal patch (Specify)	
20.	Type of package: Triple laminated Aluminum pouch	
21.	Intended Clinical Use (as in the device label/IFU-information for use)	The Drepto Mentroherb sheet is Providing relief from pain by absorbing and Transderma' contact with stomach surface.

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Annexure No 5. (Contd.,) Study Plan

Ninkilim Life Sciences Pvt Ltd

Annexure No 1. (Cont'd) Test Item Information Data Sheet (TIDS)

Ninkilim Life Sciences Pvt Ltd

22.	Test Item Categorization: 1. Surface: Intact skin 2. Contract Duration: <24 h	
23.	Test Item form	Finished form
24.	Test Item dimension (fill appropriate)	Surface area (in mm^2 /cm^2): 75 cm^2 Length (in mm/cm):15.0 cm Breadth (in mm/cm):5.0 Weight (in mg/g): About : 2.7403 g
25.	Whether the test item can be cut as needed for testing	☑ Yes ☐ No
26.	Extraction ratio to be used for testing	Not applicable
27.	Extraction condition to be used for testing	Not applicable
28.	Test Item Sterilization method adopted	Non-sterile
29.	If non-sterile, select the method of sterilization	☐ Autoclave Method ☐ Surface Sterilization ☑Tested on Non-sterile
30.	List of part(s) of the test item that should be tested (if applicable)	Not applicable
31.	A list of part(s) of the test item should be excluded from the test (if applicable)	Not applicable
Others		
32.	Particulars of dispatch: 1.Quantity and Date 2. Mode of Dispatch	 10 Pouch X 2 (20 No's) 01/08/2026 Courier
33.	Certificate of Analysis / Analytical Test Report attached	☐ Yes ☐ No ☑NA
34.	Information for use is attached	☐ Yes ☐ No ☑NA
35.	The Material Safety Data Sheet (MSDS)is attached	☐ Yes ☐ No ☑NA
36.	Hazards and precautions (if any other than those specified in MSDS or if MSDS is not provided)	Not applicable
37.	Consent to disposal of leftover test item Consent to disposal of Archived test item on expiry	Yes Yes
38.	Any additional information (optional)	Not applicable
39.	Name of the Sponsor's Representative	Rahul Kumar Gupta
40.	Name of the Monitoring Scientist	Rahul Kumar Gupta & Dr. Chandan Yadav

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Annexure No 5. (Contd.,) Study Plan

Ninkilim Life Sciences Pvt Ltd

Annexure No 1. (Cont'd) Test Item Information Data Sheet (TIDS)

Ninkilim Life Sciences Pvt Ltd

List out the test to be conducted:		
Sl. No.	**Test Title**	**Test Method**
1.	Invitro Cytotoxicity Test (DCM)	ISO 10993 Part 5:2009
2.	Skin Irritation Test (SIT-RB)	ISO 10993 Part 23:2021/Amd1:2025
3.	Skin Sensitization (BT)	ISO 10993-10:2021

Asa SponsororSponsorrepresentative ofthesestudies, I agreewithbelowpoints:

- Thestudiesrequestedare tomeet theregulatoryrequirementsofthe testitem.
- The animal usage is necessary for requested studies as per guideline requirements. The species chosen is appropriate to the study and as per guidelines requirements.
- The studies requested are not an unnecessary duplication of previous work.

SponsororSponsorRepresentative: Rahul Kumar Gupta

DREPTO BIODEVICES PVT LTD MUMBAI 400 076

Sign. and Date 07/08/25

InstructionsforfillingTestItem/ReferenceItem/Vehicle/ Placebo DataSheet:

- Fillthe informationsheetwithavailableinformation.
- Iftheinformationisnotavailablementionit asNAandifthe sectionorcolumnisnotapplicableforthe testitem,referenceitem,orvehicle,mentionit asNA(NotApplicable).

Addcolumnsor rows asperrequire

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Annexure No 5. (Contd.,) Study Plan

Ninkilim Life Sciences Pvt Ltd

Annexure No 2. Scoring System for Skin Reaction

Reaction	Irritation Score
Erythema and Eschar Formation	
No erythema	0
Very slight erythema (Barely perceptible)	1
Well-defined erythema	2
Moderate erythema	3
Severe erythema (beet-redness) to eschar formation preventing grading of erythema	4
Oedema Formation	
No oedema	0
Very slight oedema (Barely perceptible)	1
Well defined oedema (edges of area well defined by definite raising)	2
Moderate oedema (raised approximately 1 mm)	3
Severe oedema (raised more than 1 mm and extending beyond exposure area)	4
Maximal Possible Score for Irritation	**8**
Other adverse changes at the skin sites shall be recorded and reported	

Annexure No 5. (Contd.,) Study Plan

Ninkilim
Life Sciences Pvt Ltd

Annexure No 3. Primary or Cumulative Irritation Index Categories in a Rabbit

Primary Irritation Index (PII)	
Mean Score	**Response Category**
0 to 0.4	Negligible
0.5 to 1.9	Slight
2 to 4.9	Moderate
5 to 8	Severe